Growplify

Human-Computer Interaction

Phase II - Lo-fi Prototype and Heuristic Evaluation

T03G6

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Project Description

Growplify is a mobile application with the objective to help companies understand the financial impact of their operations and employees. The application uses artificial intelligence to create analyses showing whether employees, teams, or projects are generating profit for the company. From this, the application creates easy-to-understand reports that enable companies to focus on productivity and profitability, thus helping them understand areas where they can cut spending to boost profits.

The platform supports scenario modeling to assist in strategic decision-making. In other words, by manipulating different parameters of a given project, such as the number of employees and work hours per week, managers can see what the expected results are and make decisions to increase profitability.

Growplify also has features that evaluate the financial performance of partnerships, analysing each collaboration through its profit and rentability helping users to identify which can be harmful or not to the company.

Ultimately, Growplify allows companies to create value from financial data, ensuring that every operational decision and investment is used in a way to increase the value of the company.

In this second phase, we implemented three core functionalities defined in the User Analysis phase:

- Individual and team profitability analysis.
- Profit forecasting and investment simulation.
- Smart notifications.

The table below details these features and the specific user tasks designed to evaluate them in the prototype.

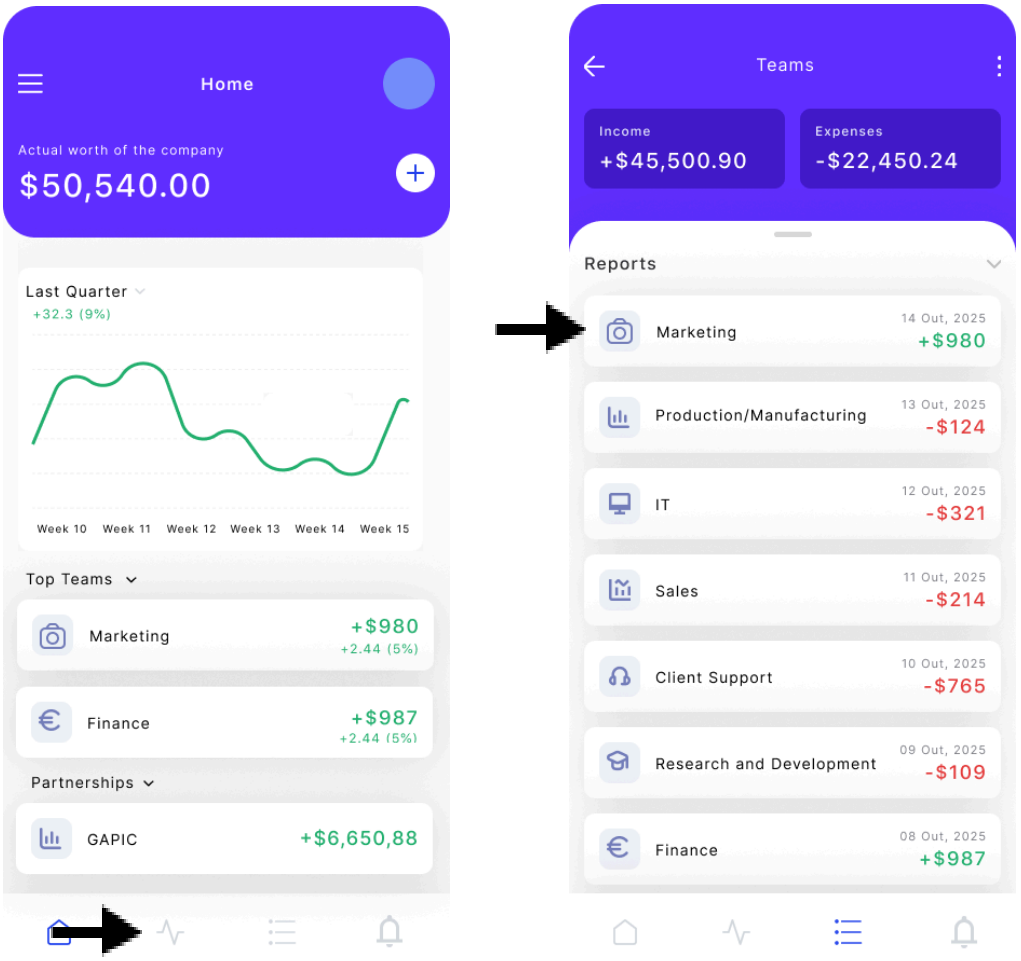
Feature	Description	Task
Individual and team profitability analysis	The user is able to select the team under analysis, as well as their members, to check the financial return, according to the allocated financial records.	I want to select the marketing team to see the financial return analysis according to the associated costs, and view the members individually.
Profit forecasting and investment simulation	The user is able to select a project and adjust project variables to instantly visualize impacts to projected profit.	I want to simulate the profit of the website development project by decreasing the time invested and increasing the number of workers.
Smart notifications	Automated, real time system that pushes notifications to users of alerts for critical deviations in financial projections, to prevent financial damage.	The marketing team's financial return fell by 8% last month, and therefore I want to receive a notification to alert me before it causes greater damage to the company.

Table 1: Developed features and tasks

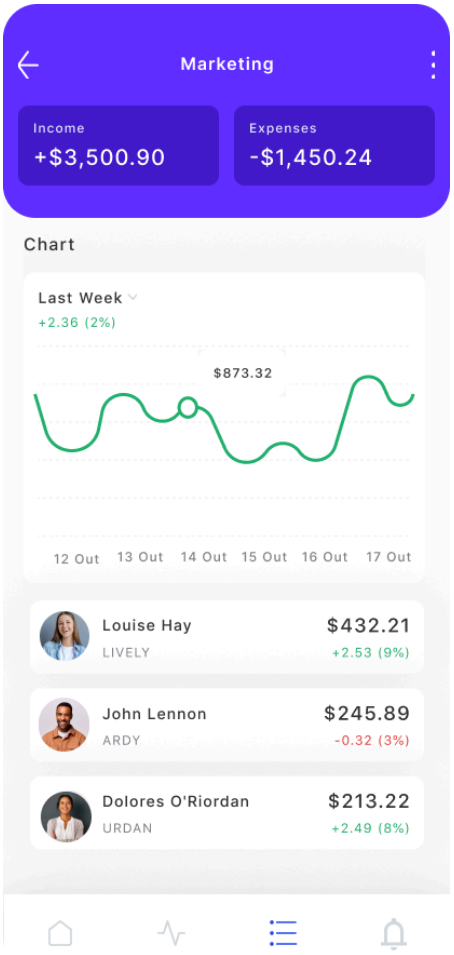
Prototype’s Wireflow

Task 1 - Individual and team profitability analysis

When accessing the homepage, the user can click on the teams button to access a new page that lists different teams.

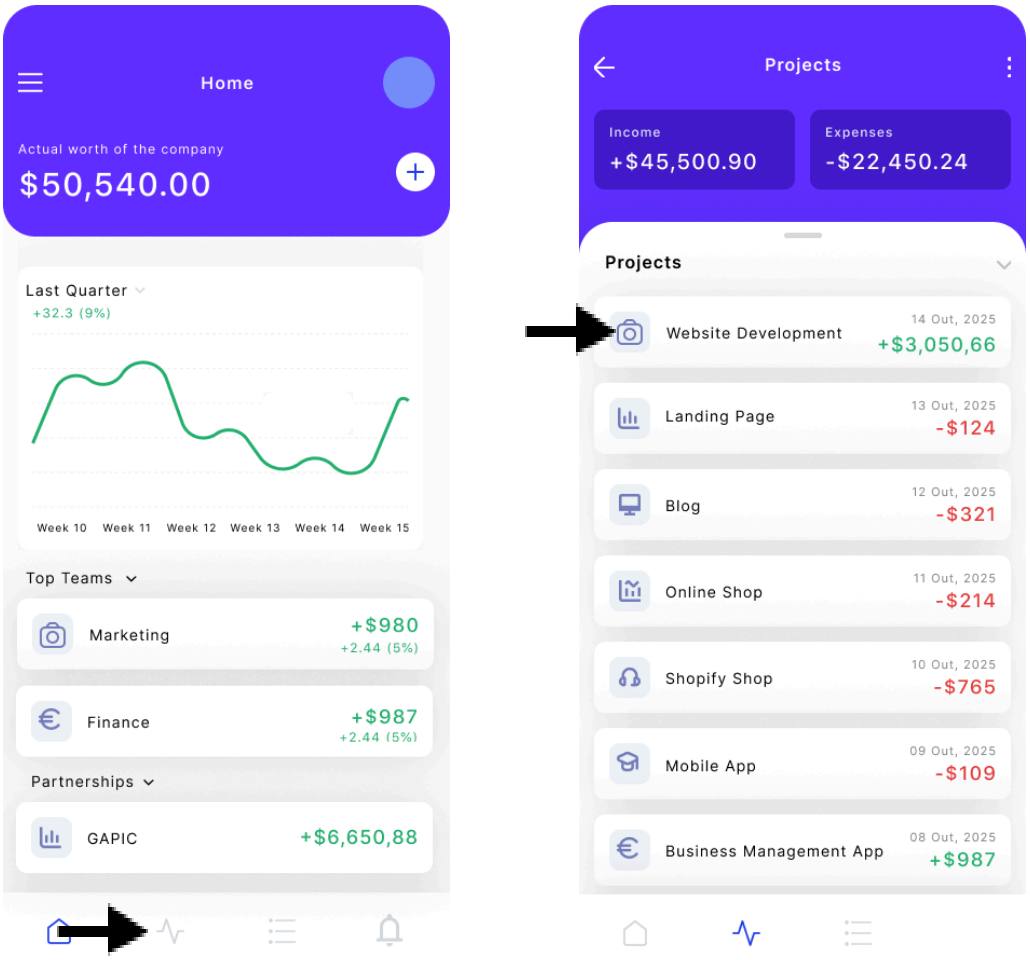


By selecting a specific team, the user will have access to the financial information of the team itself, as well as that of each of its members.

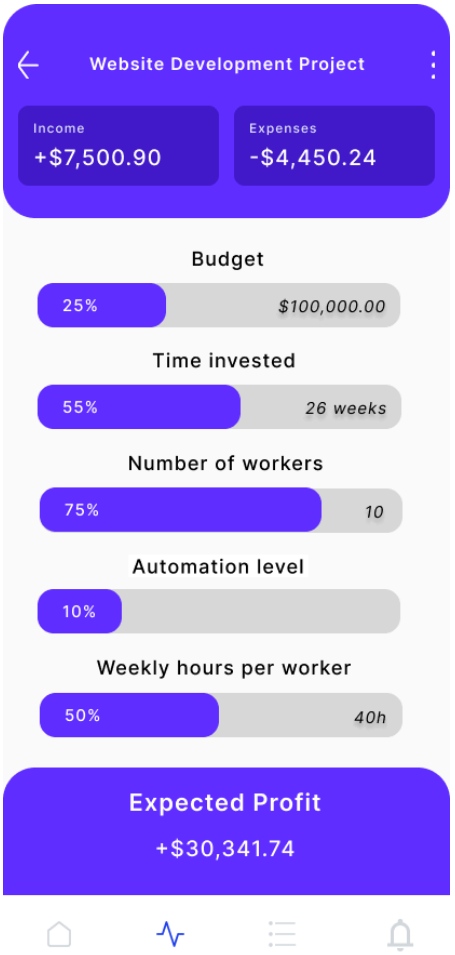


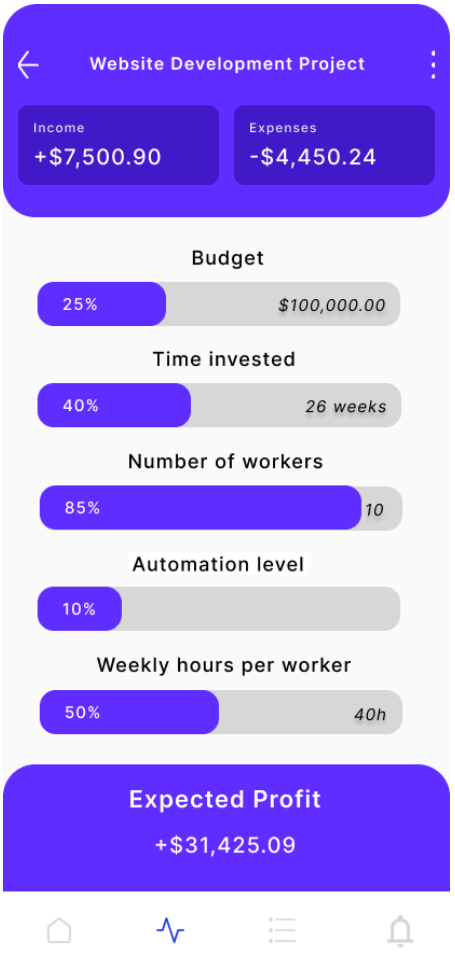
Task 2 - Profit forecasting and investment simulation

When accessing the homepage, the user can click on the projects button to access a new page that lists different projects.



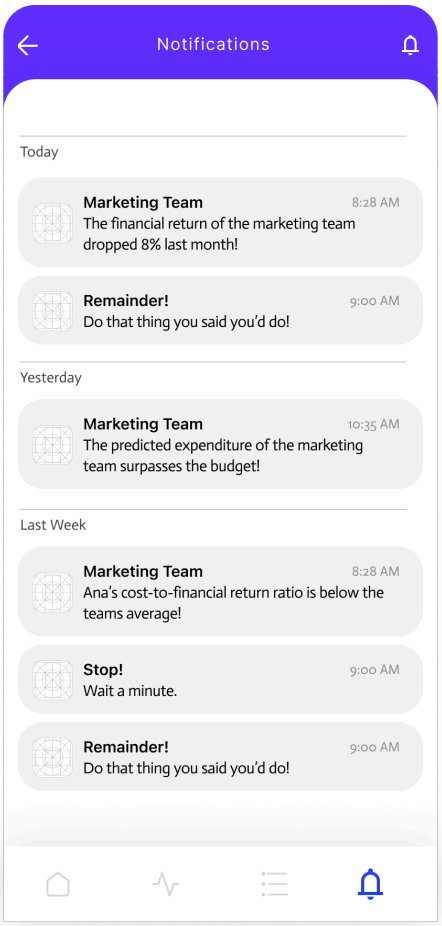
When selecting a specific project, the user will have access to a new page with different adjustable parameters. By modifying these parameters, the user can simulate investment scenarios, using artificial intelligence, to support strategic decisions.





Task 3 - Smart notifications

In a given situation, for example when the marketing team's financial return decreases by 8%, the user should receive a notification providing this information.



When selecting a notification, the user should be redirected to the notifications page where they can view them in more detail.

Digested Heuristic Evaluation Results

Taking into account the heuristic evaluation reports received, the following problems were identified:

Heuristic	Description	Severity
Visibility of system status	Graphs only show one value on the Y-axis, making it difficult to read values at other points. This makes comparing values difficult, which can lead to misinterpretations.	2
Match between system and the real world	Project icons do not clearly represent the theme of the projects.	1
User control and freedom	The user cannot manage notifications. The user may want to manage what they are notified about and when they receive it.	2
Help and documentation	There is no help section, FAQ, or support. The lack of a dedicated space to help the user can lead to difficulties in using the system, especially for new or less experienced users.	1

Table 2: Heuristic evaluation results

The discussion between the team over the identified heuristic evaluations received ended in the agreement that the feedback was non-exhaustive, occasionally focusing on the looks, which are not expected to be great in a low fidelity prototype.

Consequently, the team's attention shifted into performing our own critical analysis of the prototype, where we found a violation of the heuristic user control and freedom of severity 3:

On task 2, profit forecasting and investment simulation, the user has access to a screen where he is able to adjust parameters of a project's resource allocation, simulating different investment scenarios. However, it lacks the possibility to "reset" or "undo" the changes that have been made which limits the user and can even frustrate them when mistakes are made.

Corrections to perform in Phase 3

Taking into account the problems identified and described above, these are the corrections that are planned for the next phase:

- Add more values to the Y-axis of the chart to improve data readability. This improvement allows the user to read the graphs more accurately, making it easier to interpret the differences between points and understand the scale.

- Change the project icons to better represent the project themes, making them easier to identify. This will improve navigation and make the interface more intuitive.

- Add notification management with actions such as delete and mute. This will provide better management on the part of the user, who will gain autonomy.

- Create a Help, FAQ, or documentation section to support users. This will improve the user's learning process and reduce the number of errors made.

- We will implement a "Reset to Default" button on the simulation screen. This allows users to instantly revert all sliders to their original positions, encouraging experimentation without fear of error.

Annexes

Sent Heuristic Evaluation Reports

ipc-heuristiceval-T03-G01-G06

ipc-heuristiceval-T03-G02-G06

Received Heuristic Evaluation Reports

ipc-heuristiceval-T03-G6-G5

Avaliação Heurística — Aplicação de Estatísticas Empresariais